# Why impact also needs energy

**antocciphysics**



**Name:** _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

**LEARNING TARGET** Explain why momentum does not tell the whole impact story.

**SUCCESS CRITERIA - I can...**

1. Compare momentum and kinetic energy as different quantities.
2. Identify speed as a squared factor in kinetic energy.
3. Name where energy can go in an impact.

**Core:** teacher selects from 1-7. **Stretch:** 8 after checking your work.

**MARZANO SELF-CHECK**

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

**LANGUAGE GOAL** Use “momentum” and “kinetic energy” to explain your model or result. Sentence starters are optional.

**momentum**: mass times velocity, including direction

**kinetic energy**: energy associated with motion

**scalar**: a quantity described by size without a vector direction

## 01 Bring back an earlier idea

A and B have equal momentum magnitude, but A has 3 times B’s mass.
**Compare** speeds and kinetic energies algebraically.

**Word help:** magnitude = size of a quantity without its direction

**Sentence starter:** Both … . They differ in … because … .

**Marzano check - circle:** ① ② ③

## 02 Explain a worked example

**Reference:** $p = mv$; $K = \frac{1}{2}mv^2$.
Momentum includes direction; kinetic energy is a scalar.
Neither alone gives a complete damage forecast.

**Worked reasoning:** A 2 kg cart at 3 m/s has momentum 6 kg·m/s and kinetic energy $\frac{1}{2}\times2\times3^2=9$ J. A 1 kg cart at 6 m/s has the same momentum but 18 J of kinetic energy.

**Your turn: Draw** labeled momentum arrows and energy bars for the two carts.

**Sentence starter:** The … in my model represents … . Its label shows … .

**Marzano check - circle:** ① ② ③

## 03 Represent and interpret

Two equal-mass carts move at speeds 2 and 4 m/s.
**Predict** how their momenta and kinetic energies compare before calculating.

**Sentence starter:** Both … . They differ in … because … .



Same mass; 2 m/s  Same mass; 4 m/s

**Marzano check - circle:** ① ② ③

## 04 Solve and justify

Two objects have equal momentum 12 kg·m/s. Their masses are 2 and 6 kg.
**Find** each speed and kinetic energy.

**Sentence starter:** I found … [unit]. This value means … in this situation.

**Marzano check - circle:** ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

## 05 Analyze the reasoning

List three forms or destinations of energy after a deforming collision. Does “less kinetic energy” mean “energy destroyed”?

**Sentence starter:** I notice … . This supports … because … .

**Marzano check - circle:** 1 2 3

## 06 Choose a method

A positive triangular force pulse lasts 0.4 s and peaks at 60 N.
**Find** impulse and velocity change for a 0.8 kg object.

**Word help:** impulse = net force acting over time; change in momentum

**Sentence starter:** I found … [unit]. This value means … in this situation.

**Marzano check - circle:** 1 2 3

## 07 Exit check

At fixed momentum, mass triples. Derive or explain the kinetic-energy multiplier using $K=p^2/(2\,m)$.

**Word help:** derive = show the reasoning or algebra that produces a relationship

**Sentence starter:** Under the assumption …, the evidence … supports … because … .

**Marzano check - circle:** 1 2 3

## 08 STRETCH: Apply it another way

Two carts have equal momentum magnitude p. Derive K in terms of p and m, then compare energies if one mass is four times the other.

**Word help:** derive = show the reasoning or algebra that produces a relationship

**Sentence starter:** Both … . They differ in … because … .

**Marzano check - circle:** 1 2 3

Read down the left column, then down the right. Use the open space to draw, calculate and explain.